Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

DOVUN

Selected Domain

Domain One: Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

SDG 8 (8.5); supporting SDG 4 (4.4, 4.3)

Project / Solution Name

AIVANCIN

Name (Leader)

Jad Fahmi

Email & Phone (Leader)

jadfahmi07@gmail.com (phone in the sdchack.my submission)

University Affiliation(s)

German-Malaysian Institute

Number of Team Members

5

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Who: Malaysian graduates working below their qualifications. In Q2 2026, 40.7% of employed tertiary-educated Malaysians aged 25-34 worked in semi-skilled or low-skilled occupations: 751,100 people [1].

Moment of need: deciding whether retraining is worth their evenings and savings, and for which graduate job.

Why it is hard: a job title hides relevant work. Hafiz, our synthetic persona, is 27, holds a computing diploma and drives a delivery van. On weekends he fixes laptops and wifi, installs software, makes backups and keeps his uncle's shop website running. Search by his job title and none of the top ten roles is in computing. A course list assumes he already knows his target; chatbot advice varies between runs and is hard to check.

AI adds a second question: in his target job, which tasks should he practise doing, and which should he learn to direct and check?

Why it matters: the wrong course wastes scarce time and money and keeps skilled people in jobs below their training. Moving graduates into skilled work serves SDG targets 8.5 (decent work for young people) and 4.4 (relevant skills for employment) [2].

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AIVANCIN takes a graduate from the work they already do to the closest graduate roles and a free first plan, in three steps.
1. Describe: job, other work and studies in their own words, plus time and budget.
2. Confirm: an AI model suggests matching work activities for each sentence, from O*NET's 2,087, and highlights the words behind each. It is all-MiniLM-L6-v2, fine-tuned by the team on a RunPod GPU [3,6]. Nothing counts until the user ticks it.
3. Compare and choose: confirmed work is matched to 17 graduate roles, ranked by shared work. Up to three routes appear side by side with missing skills, entry barriers and dated evidence of change: US job projections, how often Claude already does each activity, and Malaysian group-level employment and pay trends [1,4,5]. The user picks a target and gets an ordered, free four-week plan within their time and budget, starting with one exact first task.

Scenarios (slower, moderate, faster AI adoption) show how a plan would change; in our accounting example, AI-checking practice appears in 0, 3 or 5 of 6 checkpoints. When new evidence arrives, the user decides whether to accept the change. Every role and plan step is tagged to the SDG target it serves.

Expected outcome: Hafiz leaves with three computing routes built on work he already does, what each still needs, and a free plan he can start this week. The AI suggests; the person decides.

Workshop topics, applied independently (the team did not attend): cloud hosting on Replit, structured data snapshots and model deployment without a paid API.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

CURRENT STACK (REPLIT PROTOTYPE)
Browser: HTML, CSS and JavaScript.
Backend: Python 3.12 on Replit / 3.13 locally, FastAPI and Uvicorn.
Inference: all-MiniLM-L6-v2, fine-tuned once on RunPod, quantized ONNX, CPU.
Storage: bundled CSV, JSON and NumPy snapshots; no database service. No external model API is required.

SYSTEM FLOW
Browser description -> FastAPI -> sentence encoder
-> activity suggestions -> human confirmation
-> overlap matcher + dated evidence -> routes and gaps
-> resource catalogue + constraints -> learning block.

PUBLIC DATA
O*NET 31.0, Anthropic Economic Index, DOSM, BLS outlook via O*NET OnLine, NCES and ESCO crosswalks [1,3-7]. A curated catalogue links Microsoft Learn and Saylor University resources [8].

SPONSOR USE
Replit: the live prototype, aivancin--jadfahmi.replit.app. RunPod: one RTX 4090 pod fine-tuned the encoder on O*NET task/activity pairs (US$0.42); shipped because it passed a gate fixed before training (45 vs 39 of 60 held-out). No sponsor API calls.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: DOSM data showed the scale; our insight was that a job title hides relevant work. We used synthetic personas, never real CVs.

Define: one decision for one user: which graduate role is worth working towards before paying for retraining.

Ideate: we compared job-title search, keyword search and AI suggestions. AI suggestions with confirmation won; without confirmation, 38 wrong activities entered test profiles.

Prototype: a web app for phone and desktop, live on Replit. It reviews one sentence at a time and shows entry barriers before any learning commitment.

Test: predeclared synthetic profiles (7 of 12 complete, against 2 with keywords) and rendered phone and desktop checks. Real-user testing comes next.

Deployment: the app runs on Replit from bundled, dated public data, with no paid API. Missing model files trigger a labelled keyword fallback; changed data stops startup.

Hardest risk: reading casual descriptions well in generic fields. Fallback: users confirm every suggestion, can search for missed work, and incomplete routes say so instead of guessing.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	8.5 (main): decent work for young people. AIVANCIN points graduates to skilled roles that use work they already do, with entry barriers and Malaysian pay context. 4.4: relevant skills for employment. Each missing activity on a chosen route becomes a learning goal. 4.3: affordable learning. Plan steps use free resources [2].
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	40.7% of employed tertiary-educated Malaysians aged 25-34 worked in semi- or low-skilled jobs in Q2 2026: 751,100 people [1]. Our KPI starts at 58%: 7 of 12 test profiles get a usable plan. Keyword search reaches 17% (2 of 12) and job titles 0% [9].

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Route-and-block coverage (%): the share of synthetic test profiles that get a usable plan. A usable plan means at least two relevant skilled-role routes, each with sourced entry requirements and a free four-week block covering at least three of its biggest gaps within the person's time and budget [9].
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 80% (from 58% today) on a fresh set of test profiles fixed before any changes, by the final prototype presentation [9]. Then trials with real underemployed graduates.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
  Not completed (confirmed by the team leader, 21 October 2026)
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

Narrative responses are within the word limits. Workshop completion is not claimed: the team did not attend. The last box is ticked by the team leader when signing.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☑ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Yes. Known pre-existing material:

1. KL Pro typeface, created by Jad Fahmi for DOVUN before the event (October 2026). Its unmodified files supply typography. It retains a separate, all-rights-reserved licence and is not covered by the product's Apache 2.0 licence. Distribution rights for the product were confirmed by Jad.

2. The team's README/document structure prepared before the event, adapted alongside the official SDC sample README. This is documentation structure, not pre-existing application functionality.

Third-party models, libraries, datasets, crosswalks and fonts are listed in section 6.2 and Appendix B. The submitted product code is maintained in the event repository.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET + DOSM	Workforce datasets; job outlook	CC BY 4.0
Anthropic Economic Index	Global AI-usage datasets	CC BY
all-MiniLM-L6-v2	Pre-trained sentence encoder, fine-tuned by the team	Apache 2.0
Runtime and test libraries	FastAPI, Uvicorn, ONNX Runtime, Tokenizers, NumPy, pytest, HTTPX	MIT; BSD-3-Clause; Apache 2.0; bundled notices
Crosswalks + Jost	NCES, ESCO; font. Detail: Appendix B.	Public domain; EC reuse; OFL 1.1

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

None. Jad Fahmi confirmed on 21 October 2026 that no team member has an employment, internship or consulting agreement under which an employer could claim the hackathon work.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

Jad Fahmi

[Date]

25 October 2026

Appendix A: Sources and evidence

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[1] Department of Statistics Malaysia. **Quarterly Skills-Related Underemployment by Age**, Q2 2026. Rate 40.7%; 751.1 thousand people for ages 25-34. CC BY 4.0. open.dosm.gov.my/data-catalogue/lfs_qtr_sru_age

[2] United Nations, Department of Economic and Social Affairs. SDG targets 4.3 and 4.4: sdgs.un.org/goals/goal4. Target 8.5: sdgs.un.org/goals/goal8. Descriptions in section 4 are paraphrases.

[3] USDOL/ETA. **O*NET 31.0 Database**, files dated 25 August 2026; CC BY 4.0. onetcenter.org/database.html. AIVANCIN uses modified extracts. USDOL/ETA has not approved, endorsed or tested these modifications. O*NET is a trademark of USDOL/ETA.

[4] Anthropic. **Economic Index**, bundled releases 15 September 2025, 15 January 2026, 24 March 2026 and 26 June 2026. CC BY release notices. anthropic.com/economic-index. Usage describes global Claude conversations, not Malaysian labour demand.

[5] O*NET OnLine, BLS US 2025-2035 job projections: onetonline.org/find/bright. Malaysian observed employment/pay: open.dosm.gov.my/data-catalogue/wages_occupation. US projections and Malaysian group observations remain separate.

[6] sentence-transformers. **all-MiniLM-L6-v2**, pinned revision 1110a243, quantized ONNX. Apache 2.0. huggingface.co/sentence-transformers/all-MiniLM-L6-v2.

[7] NCES, CIP 2020 to SOC 2018 crosswalk (US federal work): nces.ed.gov/ipeds/cipcode. European Commission, ESCO O*NET-ISCO crosswalk; reuse with attribution under Decision 2011/833/EU: esco.ec.europa.eu.

[8] Linked learning providers: Microsoft Learn and Saylor University. Links and provider objectives are curated; missing coverage is explicit. Resource availability is checked when catalogued, not guaranteed indefinitely.

[9] Team offline evaluation, 23 October 2026 (python tools/checks/fresh_eval.py, set 8). Predeclared synthetic profiles: first run 7/12 in scope meet the route-and-block bar (58.3%); title-only 0/12, BM25 keyword 2/12. Out of scope 4/4; no invalid blocks or errors. After the negation repair: 7/12 again, 31/57 true activities confirmed versus 28/57 first run, 21 relevant and 5 off-family routes. This rerun is not independent. Target: at least 80% on a fresh predeclared set. No real-user benefit is established. Full checks and limitations: repository evidence. Visual documentation: aivancin.vercel.app.

Appendix B: Resource and licence detail

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Runtime code: FastAPI and ONNX Runtime, MIT; Uvicorn and NumPy, BSD-3-Clause; Tokenizers, Apache 2.0. NumPy also contains bundled components under 0BSD, MIT, Zlib and CC0 notices. Those notices remain applicable.

Test code: pytest, MIT; HTTPX, BSD-3-Clause. These support offline verification and are not external product APIs.

Model: all-MiniLM-L6-v2 from sentence-transformers, Apache 2.0, fine-tuned once by the team on a RunPod RTX 4090 with O*NET 31.0 task-to-activity pairs, then quantized to ONNX. It shipped because it passed a gate fixed before training: 45 of 60 held-out descriptions against 39 for the original model.

Datasets: O*NET 31.0, O*NET OnLine outlook extracts and DOSM labour/pay datasets, CC BY 4.0; Anthropic Economic Index, CC BY as recorded in each release. Source releases, access dates, modifications and hashes are recorded in the repository data manifest.

Crosswalks: NCES CIP-SOC, US federal government work (public domain); European Commission ESCO O*NET-ISCO mapping, attribution and reuse under Commission Decision 2011/833/EU.

Fonts: Jost by Owen Earl and the Jost Project Authors, SIL Open Font License 1.1. KL Pro by Jad Fahmi for DOVUN, pre-existing, all rights reserved, separate licence. KL Pro is not included in the project Apache 2.0 grant.

Documentation structure: official SDC sample README structure, adapted alongside the pre-event team README structure. The official sample repository is distributed under Apache 2.0. No sample application functionality is claimed as original AIVANCIN functionality.

Course resources: external links and factual metadata, rather than copied course content. Microsoft material remains subject to its terms. Saylor-authored content is CC BY 3.0, excluding final exams and separately licensed third-party content. AIVANCIN does not redistribute course lessons.

Current limitations: English intake; US occupation and requirement data; Malaysian pay/employment at broad group level; weak route coverage in generic fields. Moderate AI-use outlook backtest MAE: 16.33 percentage points. Slower/faster scenarios are assumptions. They are not Malaysian employment forecasts.

Team status: the leader supplied contact details and confirmed no workshop attendance and no employment-related IP agreements on 21 October 2026. Workshop completion is left unticked. Sponsor use (Replit deployment, RunPod fine-tuning) is declared, with proof, in the Technical and Tools Declaration.